Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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1 Introduction

Use XeLaTex to compile please.

1.1 Table

Table 1 gives an example.

All tables, figures, and equations must be capitalized when cited (e.g., As shown in Table 1, ...).

All tables, figures, and equations included must be cited correctly.

All tables and figures must have a caption.

Tabel 1: Example table

Parameter	Value	Description
a	1	one
b	2	two
c	3	three

1.2 Image

Figure 1 gives an example.

1.3 Equation

Equation eq:first is an example.

$$\frac{\partial u(t,x)}{\partial t} = Au(t,x)\left(1-\frac{u(t,x)}{K}\right) - B\frac{u(t-\tau,x)w(t,x)}{1+Eu(t-\tau,x)}, \frac{\partial w(t,x)}{\partial t} = \delta\frac{\partial^2 w(t,x)}{\partial x^2} - Cw(t,x) + D\frac{u(t-\tau,x)w(t,x)}{1+Eu(t-\tau,x)}. \tag{1}$$

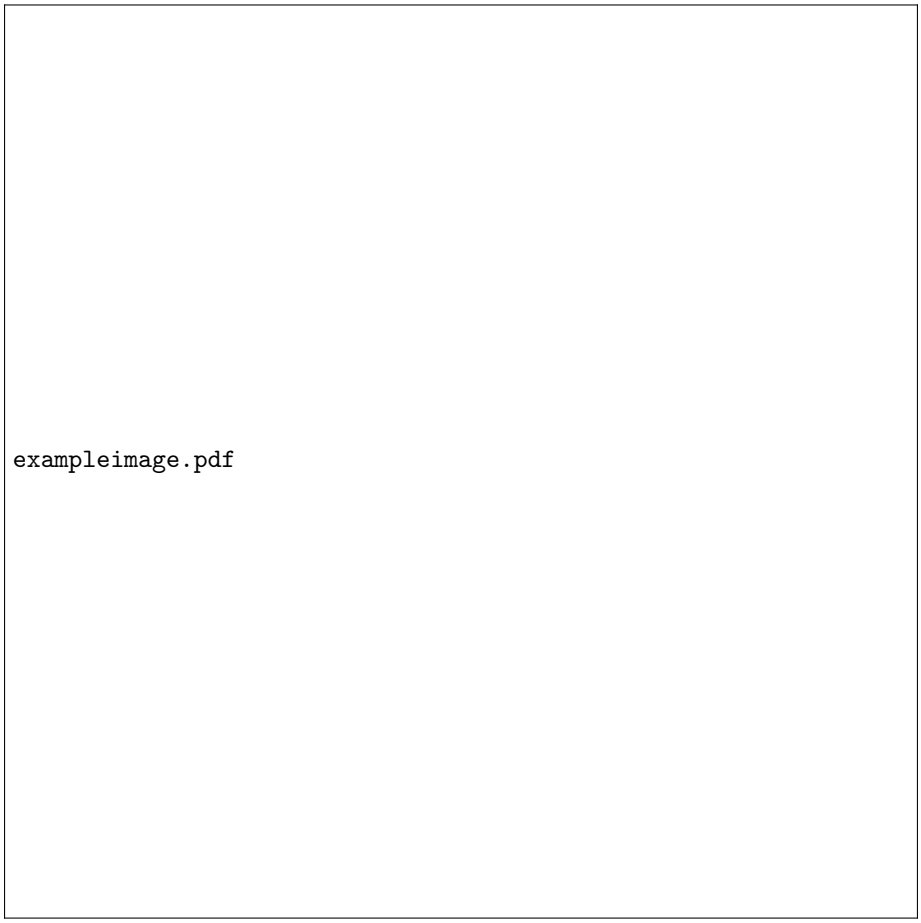
1.4 Citation

Titles of all cited articles are required. Titles of articles cited in reference lists should be in upright, not italic text; the first word of the title is capitalized, the title written exactly as it appears in the work cited.

Author names are listed in Roman alphabet characters as standard. We can support presentation of author names using some non-Roman characters in parentheses after the Roman-character spelling.

Cite this way [**?**].

Cite this way if the original source is in non-Roman alphabet, such as Chinese [**?**]. Author names in their original characters should follow the Roman alphabet characters, and the same applies to the title and the name of the journal or book. If any characters are unsupported, use only the Roman alphabet characters.



Gambar 1: This is an example image

Acknowledgement

Insert the Acknowledgment text here.